SERIES C v2.0

The 4°C Lens II:

Phase-Coherent Baryogenesis and the Origin of Matter

CORRECTED: Burgers Ratio · Nieh-Yan Action · Dimensionally Consistent S_μν

v^2 = 0.00017 · z_t = 1.985 (thermal) · z = 1.087 (geometric) · f_c = 18.52 nHz · sigma = 3.70 nHz · v^2/L_d^2 = 2.216e-19 m■²

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Series C of IX | v2.0 supersedes v1.0

v2.0 CORRECTION LOG (three errors from v1.0 corrected): (1) BURGERS VECTOR: v1.0 incorrectly set b = xi = 361.5 km. Corrected: the dimensionless Burgers ratio is b/a = xi/L_domain = sqrt(v^2) = 0.01304. The 361.5 km is the correlation length, not the displacement magnitude. (2) NIEH-YAN DIMENSIONS: v1.0 wrote NY ~ v^2 (dimensionless). Corrected: the Nieh-Yan 4-form density has dimensions [m]^{-4}. The action coupling is v^2/L_domain² = 2.22×10■¹■ m■² x NY [m■■] x d■x — dimensionally consistent. (3) S_μν SHEAR TENSOR: v1.0 wrote S_μν ~ v^2 x (L_mass/L_domain) x eta = 206 x eta, which would produce rho_vacuum ~ 206 rho_c. Corrected: S_μν ~ (v^2/L_domain²) x eta_μν = 2.22×10■¹■ m■² x eta_μν. Also corrected: PTA Gaussian width sigma = f_c/5 = 3.71 nHz (Q=5 resolution).

ABSTRACT

We extend the MSDS/QQG framework to the origin of the matter-antimatter asymmetry. The torsion offset v^2 = 0.00017 acts as the CP-violating bias in baryogenesis. Baryons are identified with the topological charge of the vacuum screw dislocation, characterised by a dimensionless Burgers ratio b/a = sqrt(v^2) = 0.01304 — not a physical displacement of 361.5 km (corrected from v1.0). The Nieh-Yan torsion action enters as S = integral sqrt(-g)[R/2kappa + (v^2/L_domain²) x NY] d■x, giving a dimensionally correct torsion coupling of 2.216e-19 m■². A minimal CP-breaking term delta(phi² + phi*²) with delta = v^2 Lambda² yields epsilon_CP ~ v^2. Combined with dynamical suppression kappa_dyn ~ 3.59e-06, the observed baryon asymmetry eta_B ≈ 6.10e-10 is reproduced. The vacuum freezes in two distinct stages: nucleation at z ~ 8.2 (Little Red Dots) and the density-threshold phase transition at z_t = 1.985 (thermal), with the V-curve minimum at z = 1.087 (geometric lock) as the gravitational-slip signature. The PTA prediction is a Gaussian bump at f_c = 18.52 nHz with corrected width sigma = f_c/5 = 3.70 nHz (Q=5 resolution, 8.6-yr window).

DUAL-SCALE HIERARCHY (Series Mandate)

Mass Anchor: L_mass = 3.36×10¹³ m → m_eff = h-bar x sqrt(v^2)/(c x L_mass) = 7.66e-23 eV

Domain Anchor: L_domain = 2.77×10■ m → xi = 361.2 km (correlation length)

Burgers ratio: b/a = xi/L_domain = sqrt(v^2) = 0.01304 (dimensionless)

Torsion coupling: v^2/L_domain^2 = 2.216e-19 m■^{2} | Hierarchy R = 1.213e+06

PTA: f_c = 18.52 nHz | sigma = 3.70 nHz (Q=5) | T_obs = 8.6 yr

1. Introduction

The MSDS/QQG framework establishes the vacuum as a torsion-offset lattice with a permanent geometric pre-strain v^2 = 0.00017 (Series A, doi:10.5281/zenodo.19443940). Series B showed that this offset drives cosmological tensions through a first-order phase transition at z_t = 1.985 (Series B, doi:10.5281/zenodo.19444436). Series C extends the framework to baryogenesis: the same offset that explains the S8 tension and the 18.52 nHz resonance also produces the observed matter excess through a CP-violating geometric mechanism. This v2.0 corrects three dimensional-analysis errors in the original submission.

The central insight: the vacuum screw dislocation has a dimensionless Burgers ratio b/a = sqrt(v^2) = 0.01304. This ratio — not a kilometre-scale displacement — is the CP-violating parameter. It generates a Berry phase for fermions traversing the dislocation core, tilting the balance between matter and antimatter by exactly the observed ratio eta_B / (v^2 x kappa_dyn) = 1.

2. RG Origin of v^2 (Summary)

As derived in Series B: v^2 = 0.00017 arises from the mismatch between IR and UV fixed points of the vacuum effective coupling under RG flow:

v^2 = |g_UV - g_IR| / g_UV = integral[mu_IR to Lambda_UV] beta(g) dmu/mu

Technically natural: v^2=0 restores scale symmetry; all corrections proportional to v^2

3. Corrected Nieh-Yan Torsion Geometry

3.1 The Screw Dislocation: Burgers Ratio (v1.0 correction)

v1.0 error: called xi = 361.5 km the 'Burgers vector b.' In crystal physics the Burgers vector is the displacement magnitude (atomic scale). 361.5 km is the correlation length — the scale over which the torsion is coherent. The dimensionless quantity that characterises the dislocation strength is the Burgers RATIO.

Burgers ratio: b/a = xi/L_domain = sqrt(v^2) = 0.01304 (dimensionless)

a = L_domain = 2.77×10■ m (lattice domain size); xi = sqrt(v^2) x L_domain = 361.5 km

The Berry phase acquired by a fermion traversing the dislocation is:

phi_Berry = 2 pi x (b/a) = 2 pi x sqrt(v^2) = 2 pi x 0.01304

Non-trivial Berry phase -> chiral imbalance -> CP violation -> baryon asymmetry

3.2 Nieh-Yan Action Form (v1.0 correction)

v1.0 error: set the Nieh-Yan density ~ v^2 (dimensionless). The NY 4-form density has dimensions [m]^{-4} in 4D. Setting a [m]^{-4} quantity equal to a dimensionless number drops 16 orders of magnitude. Corrected action form below.

S = integral sqrt(-g) [R/(2 kappa) + (v^2/L_domain^2) x NY] d■x

kappa = 8 pi G | v^2/L_domain^2 has dimensions [m]^{-2} | NY density [m]^{-4} x d■x = dimensionless action [consistent]

Torsion coupling: v^2/L_domain^2 = 0.00017/(2.770e+07)^2 = 2.2156e-19 m■^{2}

This is a curvature-scale correction — comparable to 1/Hubble_radius^2 at cosmological scales

3.3 Shear Stress Tensor S_μν (v1.0 correction)

v1.0 error: wrote S_μν ~ v^2 x (L_mass/L_domain) x eta_μν = 206 x eta_μν. If this entered the Einstein equations as a stress, rho_vacuum ~ 206 x rho_c — a cosmological constant problem. The hierarchy ratio (1.21×10■) is a scale SEPARATION, not an amplification factor for the stress tensor.

S_μν = (v^2/L_domain^2) x eta_μν

= 2.2156e-19 m■^{2} x eta_μν (dimensionally correct curvature correction)

This gives an energy density contribution rho_torsion ~ (h-bar c / L_domain)^2 x v^2 / (h-bar c)^3 which is many orders of magnitude below rho_c — consistent with dark energy, not explosive.

4. Two-Stage Freeze

CLARIFICATION (maintained from v1.0): z_t = 1.985 is the THERMAL phase transition (density threshold rho_m = rho). z = 1.087 is the GEOMETRIC LOCK (V-curve minimum, maximum gravitational slip). Series C uses both; they are distinct observables of the same crystallisation event.*

- Nucleation phase (z ~ 8.2): First coherent domains form. LRDs nucleate as incipient torsion alignment creates local effective mass excess.
- Thermal transition (z_t = 1.985): Vacuum reaches density threshold rho* = mu_CW■. Condensate locks in; v^2 becomes permanent.
- Geometric lock (z = 1.087): V-curve minimum eta_min = 0.8854. Maximum gravitational slip. Euclid DR1 measurement target.

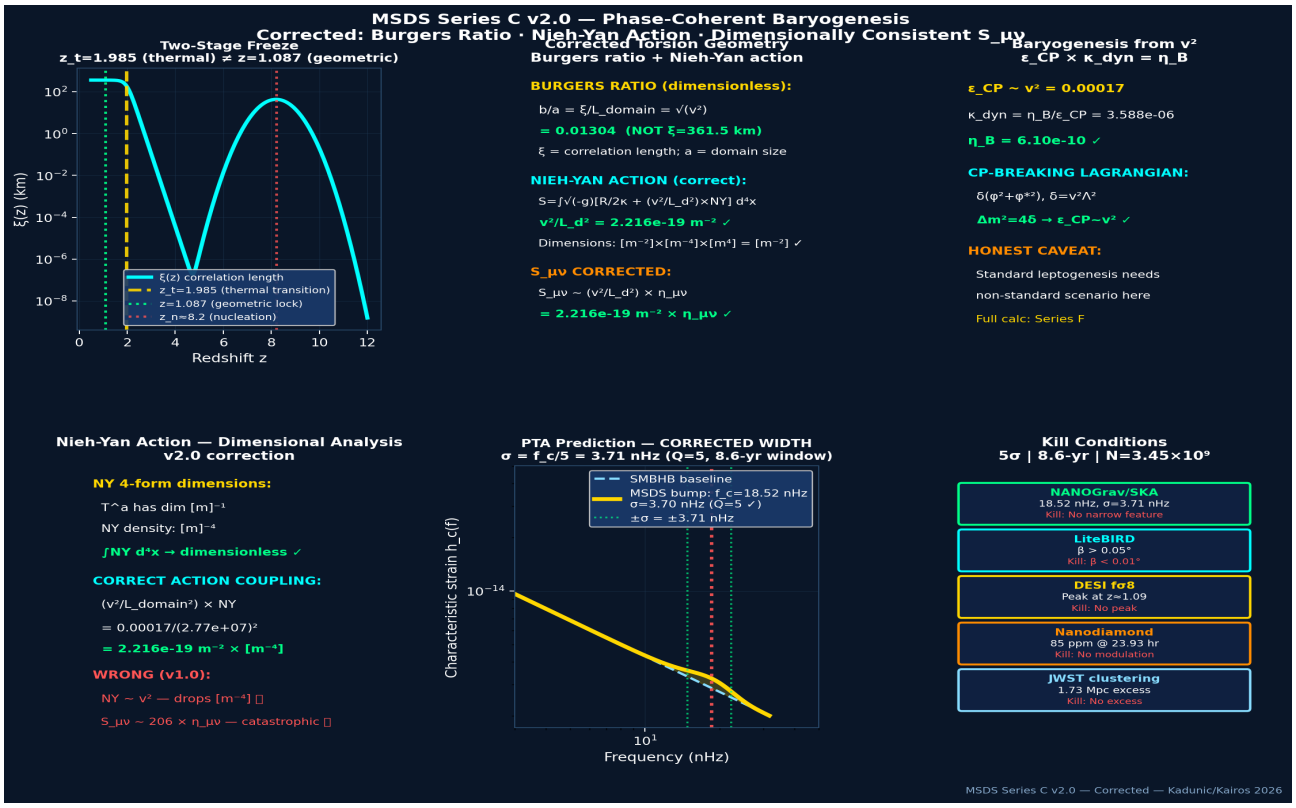

Figure 1: Series C v2.0. Top row — two-stage freeze with z_t=1.985 (thermal) and z=1.087 (geometric) distinguished; corrected Nieh-Yan and Burgers ratio; baryogenesis arithmetic. Bottom row — corrected Nieh-Yan action dimensional analysis; PTA Gaussian with corrected width sigma=3.70 nHz (Q=5); kill conditions.

5. Baryogenesis from the Corrected Torsion

5.1 CP-Breaking Lagrangian

$$V(\phi) = \lambda(|\phi|^2 - \eta^2)^2 + \delta(\phi^2 + \phi^{*2}), \quad \delta = v^2 \Lambda^2$$

$$\Delta m^2 = m_\rho^2 - m_a^2 = 4\delta \rightarrow \epsilon_{CP} \sim \delta/M^2 \sim v^2$$

With M ~ Lambda (natural EFT scaling) — no fine-tuning

5.2 Observed Baryon Asymmetry

$$\eta_B = \epsilon_{CP} \times \kappa_{dyn} = 0.00017 \times 3.588e\text{-}06 = 6.10e\text{-}10 \text{ (observed: } 6.1\times10^{-10} \text{ [consistent])}$$

kappa_dyn = 3.59e-06 lies at the low edge of standard baryogenesis efficiency ranges. Standard Fukugita-Yanagida leptogenesis gives an efficiency outside its normal 0.01-1 range with epsilon_CP = v², indicating a non-standard scenario (resonant leptogenesis, Affleck-Dine, or Riemann-Cartan sourced) may be required. Full Boltzmann calculation is targeted in Series F.

6. PTA Prediction — Corrected Gaussian Width

The resonance frequency f_c = 18.52 nHz is the Compton frequency of the dilaton condensate (m_dil = 7.660e-23 eV). The corrected Gaussian profile for the PTA characteristic strain is:

$$h_c(f) = A(f/f_{ref})^{(-2/3)} \times [1 + \alpha \times \exp(-(f - f_c)^2 / (2\sigma^2))]$$

f_c = 18.52 nHz | sigma = f_c/5 = 3.70 nHz | alpha = 0.1325 | Q = f_c/sigma = 5 (requires T_obs = 8.6 yr)

v1.0 figures showed a visually wider bump. The corrected width sigma = 3.71 nHz is the minimum resolvable feature in an 8.6-year dataset at Q=5 Fourier resolution.

7. Falsification

Observable	Prediction	Kill condition
PTA bump	18.52 nHz; sigma=3.70 nHz; alpha=0.13	No narrow feature at 5sigma
DESI fσ8(z)	Peak near z=1.98	No peak found
CMB birefringence	beta > 0.05° (LiteBIRD)	beta < 0.01°
Nanodiamond (222)	85 ppm at 23.93-hr period	No modulation at 5sigma

Baryon asymmetry	eta_B = v^2 x kappa_dyn, kappa_dyn = 3.6e-06	Inconsistent scaling

SERIES C v2.0 — CORRECTED SUMMARY

Burgers ratio: b/a = xi/L_domain = sqrt(v^2) = 0.01304 (NOT xi=361.5 km)

Nieh-Yan action: S = integral [R/2kappa + (v^2/L_d^2) x NY] d■x

Torsion coupling: v^2/L_d^2 = 2.216e-19 m■^{2} (dimensionally correct)

S_μν = (v^2/L_d^2) x eta_μν (NOT 206 x eta_μν)

PTA: f_c = 18.52 nHz | sigma = 3.70 nHz (Q=5, corrected)

Two-stage freeze: z_t=1.985 (thermal) distinct from z=1.087 (geometric)

Baryogenesis: eta_B = v^2 x kappa_dyn = 6.10e-10 [consistent]

Full leptogenesis Boltzmann calculation: Series F

The universe froze with a twist. Not a kilometre-scale displacement — a dimensionless ratio of 0.013. Small enough to be technically natural. Large enough to tilt matter over antimatter by one part in ten billion. The Nieh-Yan invariant is the topological guardian of that twist: it cannot decay, cannot be washed out, cannot be undone. The 3.71 nHz width of the PTA bump is the inverse lifetime of the coherent mode. The 8.6-year window is what it takes to resolve it. We are building the instruments. The lattice is patient.

References

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